Hotel Review Sentiment & Aspect Rating Analysis

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1. Problem Statement

Hotel reviews contain detailed information about customer experience, but manually analyzing large numbers of reviews is time-consuming. This project investigates whether traditional Natural Language Processing (NLP) and Machine Learning techniques can automatically predict the overall hotel rating from review text, convert the rating into sentiment polarity, and model ratings for individual hotel aspects.

The project also compares different text representations, learning algorithms, and class-balancing strategies to understand their effect on prediction performance.

2. Dataset

The experiment uses a 1,000-review English sample from a processed TripAdvisor hotel-review dataset. The title and body of each review are combined into a single review-text input. Valid ratings are restricted to the 1–5 range.

Item	Details
Number of reviews	1,000
Input	Review title + review body
Overall target	1–5 stars
Aspect targets	Rooms, Location, Value, Cleanliness, Sleep Quality
Train/Test split	Approximately 2/3 training, 1/3 testing
Random state	42

The available aspect labels are *rooms*, *location*, *value*, *cleanliness*, and *sleep quality*. The working dataset is a smaller substitute dataset and therefore represents an approximation rather than an exact reproduction of the original reference study.

3. Methodology and Implementation

The modelling pipeline is:

Review Text → Preprocessing → Vectorization → Sparse Features → ML Model → Prediction → Evaluation

The project compares three main text representations:

- **Word unigrams:** individual words.
- **Word 2/3-grams:** two- and three-word sequences that capture short phrases and word order.
- **Character 5-grams:** five-character sequences that can capture subword and spelling patterns.

Both count-based and TF-IDF representations are evaluated.

The machine-learning algorithms are:

- Multinomial Naive Bayes
- Linear Support Vector Machine (LinearSVC)

- Linear Regression

The Linear SVM uses $C = 1$ and `max_iter=5000`.

For sentiment evaluation, numerical ratings are mapped as follows:

Rating	Polarity
1–2	Negative
3	Neutral
4–5	Positive

Class imbalance is evaluated using four strategies: no balancing, inverse-frequency class weighting, random oversampling, and random undersampling.

4. Experimental Results

4.1 Algorithm Comparison

Algorithm	Star Accuracy	Polarity Accuracy
Multinomial Naive Bayes	48.8%	78.7%
Linear SVM	**57.2%**	80.5%
Linear Regression	53.6%	80.8%

Linear SVM provides the strongest exact 1–5 star result in the direct algorithm comparison, with an accuracy of 57.2%.

4.2 Class-Balancing Experiment

Balancing Strategy	Star Accuracy
No balancing	55.4%
Class weighting	**57.2%**
Oversampling	55.7%
Undersampling	47.9%

Class weighting performs best among the tested balancing strategies.

4.3 Feature Representation Comparison

The highest exact-star accuracy is 57.2%, achieved by both unigram TF-IDF and character 5-gram TF-IDF. The highest polarity accuracy is 81.1%, achieved by character 5-gram TF-IDF.

5. Aspect-Level Analysis

The project also compares models trained using overall ratings with models trained directly on individual aspect ratings.

Aspect-specific training improves performance for Location and Cleanliness in this experiment, while overall-trained models perform better for Rooms, Value, and Sleep Quality. Therefore, aspect-specific training is not automatically better for every target.

6. Main Findings

The main findings are:

Representation	Star Accuracy	Polarity Accuracy
1-word counts	54.8%	80.5%
1-word TF-IDF	**57.2%**	80.5%
5-character counts	53.9%	79.3%
5-character TF-IDF	**57.2%**	**81.1%**
2/3-word counts	53.9%	79.3%
2/3-word TF-IDF	54.5%	79.0%

Aspect	Overall-trained	Aspect-trained
Rooms	53.6%	47.3%
Location	56.3%	**63.8%**
Value	**46.1%**	42.8%
Cleanliness	59.3%	**61.1%**
Sleep Quality	**54.2%**	52.1%

1. Linear SVM gives the strongest exact 1–5 star result in the direct algorithm comparison at 57.2%.
2. Class weighting is the strongest tested balancing strategy, also reaching 57.2% star accuracy.
3. Feature representation materially affects performance.
4. The highest exact-star accuracy in the feature comparison is 57.2%.
5. Character 5-gram TF-IDF gives the highest polarity accuracy at 81.1%.
6. Aspect-specific training helps some aspects but not all.

The final paper-style configuration uses a class-weighted Linear SVM with word 2/3-gram count features. This should not be confused with the separate feature experiment: the 81.1% polarity score belongs specifically to character 5-gram TF-IDF.

7. Demonstration

For the demonstration review *"Convenient stay"*, the model predictions were:

Target	Predicted Rating	Polarity
Overall	4	Positive
Rooms	5	Positive
Location	5	Positive
Value	3	Neutral
Cleanliness	5	Positive
Sleep Quality	4	Positive

8. Conclusion

The project demonstrates an end-to-end traditional NLP machine-learning pipeline for hotel-review analysis. It compares multiple text representations, algorithms, and class-balancing strategies and extends the analysis to hotel-specific aspects.

The experiments show that model performance depends strongly on feature representation. Linear SVM provides the strongest exact-star result in the direct algorithm comparison, while class weighting performs best among the tested balancing strategies. Character 5-gram TF-IDF produces the highest polarity accuracy in the feature comparison.

9. Limitations and Future Scope

The working dataset contains only 1,000 reviews, so the work is an approximation rather than an exact reproduction of the much larger reference study. The substitute dataset also does not contain every

original aspect. A random review-level split may place reviews from the same hotel in both training and testing partitions.

Future work can include:

- using a larger hotel-review dataset,
- performing hotel-level train/test splitting,
- hyperparameter tuning,
- richer text preprocessing,
- testing transformer-based models such as BERT.

10. Submission Alignment

The Machine Learning mini-project guidelines require a private GitHub repository containing the source code and a README with setup and run instructions. The write-up must be submitted as a concise two-page PDF covering the problem statement, dataset details, approach, implementation, and conclusions. The final review also requires a presentation and a live working-code demonstration.